# CHAPTER V

# Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

### 5.1. Intellectual Property Considerations

Intellectual property (IP) considerations involve understanding and addressing issues related to the ownership, protection, and use of intellectual assets.

**Example:** Imagine you're developing a novel algorithm for secure data encryption in cloud computing. In this case, you need to consider who owns the rights to the algorithm—whether it's your academic institution, a funding agency, or yourself. If your work has the potential for commercial applications, you might want to explore patenting aspects of the algorithm to protect your intellectual property.

### 5.2. Ethical Considerations

Ethical considerations involve ensuring that your project is conducted in an ethical and responsible manner.

**Example:** Suppose your project involves collecting user data for training a machine learning model. Ethical considerations would include obtaining informed consent from users, ensuring data privacy and confidentiality, and being transparent about how the collected data will be used. If your project involves potentially controversial applications, such as facial recognition technology, careful consideration of the ethical implications and societal impacts is crucial.

### 5.3. Safety Considerations

Safety considerations involve identifying and mitigating potential risks to the well-being of individuals or the environment. In a computer science project, safety considerations may include:

**Example:** If you are developing software for autonomous vehicles, safety considerations would involve rigorous testing to ensure the reliability and robustness of the software. It also involves implementing fail-safe mechanisms to prevent accidents in case of software or hardware failures.

### 5.4. Legal Considerations

Legal considerations encompass adherence to laws and regulations relevant to your project.

**Example:** Suppose your project involves the development of a healthcare application that handles sensitive patient data. Legal considerations would include compliance with data protection laws such as HIPAA (Health Insurance Portability and Accountability Act) in the United States or GDPR (General Data Protection Regulation) in Europe. Ensuring that your project adheres to relevant legal frameworks is essential to avoid legal consequences.

Addressing these considerations not only ensures the responsible conduct of your project but also contributes to its overall success and acceptance within the academic and professional communities. It's essential to integrate these considerations into your project planning and execution from the outset to navigate potential challenges and create a well-rounded and impactful contribution to the field of computer science.

## 5.5. Impact of the Project on Societal, Health, and Cultural Issues

The impact of your project on societal, health, and cultural issues refers to how your thesis work contributes to addressing or improving aspects of society, health, and culture.

***5.5.1. Societal Impact:*** Suppose your project involves developing a mobile application aimed at improving access to education for underprivileged communities. By providing free educational resources and tools, your project directly addresses societal issues related to education inequality and empowers individuals with knowledge and skills they might not otherwise have access to.

***5.5.2. Health Impact:*** Consider a project that focuses on developing a smart healthcare monitoring system for elderly individuals living alone. This system tracks vital signs, detects anomalies, and alerts caregivers or medical professionals in case of emergencies. Such a project can significantly impact health outcomes by enabling timely intervention and reducing the risk of adverse health events for vulnerable populations.

***5.5.3. Cultural Impact:*** Imagine a project that revolves around preserving indigenous languages through digital platforms and interactive learning tools. By creating resources that promote language retention and cultural appreciation, your project helps safeguard cultural heritage and fosters a sense of identity and belonging among indigenous communities.

## 5.6. Impact of Project on the Environment and Sustainability

The impact of your project/thesis on the environment and sustainability relates to how your thesis work addresses environmental challenges or promotes sustainable practices.

***5.6.1. Environmental Impact:*** Suppose your project focuses on optimizing energy consumption in smart buildings through IoT (Internet of Things) technology and machine learning algorithms. By developing algorithms that regulate heating, cooling, and lighting systems based on occupancy and environmental conditions, your project reduces energy waste and minimizes the carbon footprint of buildings, contributing to environmental conservation efforts.

***5.6.2. Sustainability Impact:*** Consider a project that explores innovative materials and construction techniques for eco-friendly housing solutions. By experimenting with sustainable building materials such as bamboo, recycled materials, or low-impact construction methods, your project promotes sustainable practices in the construction industry and offers affordable, environmentally friendly housing options that minimize resource depletion and environmental degradation.

***5.6.3. Waste Reduction:*** Another example could involve developing a waste management system that utilizes data analytics and optimization algorithms to optimize waste collection routes, reduce landfill overflow, and promote recycling initiatives. By streamlining waste management processes and promoting recycling practices, your project contributes to reducing environmental pollution and conserving natural resources.

# Chapter I

## Write about work distribution of this project/thesis if it is done by two or more students

This thesis/project is done by three students namely Sonia, Karim & Dulal.

**Sonia** worked on the C code running on the PIC32, including updates to support 2 predators over serial read, updates to the boids animation to reflect predators eating them and scoring points, and direct digital synthesis sound effects. She also implemented the C++ aspects of communication option 3, including the final test implementation with the SFML bouncing ball library and WiringPi serial read. She also collaborated with Karim on coding the TFT game elements to optimize the gameplay experience, such as the scoring and boundary displays, the winning player screen, as well as other attempts at gameplay extensions like a start screen which did not make it into the final implementation.

**Karim** developed the computer vision system, consisting of Python scripts running on a Raspberry Pi. The system consisted of several stages: video manipulation, image processing, object detection, and centroid estimation. He wrote the code for each stage iteratively, optimizing the sequence and parameters of operations to produce reliable and accurate tracking of two laser dots simultaneously in a range of lighting conditions. He also developed a calibration program that allows quick tuning of image processing parameters to specific indoor environments. In addition, he collaborated with Sonia and Dulal on developing the various iterations of serial communication on the Pi, and was responsible for color tuning and computer vision setup during full system testing in various classrooms.

**Dulal** wrote the code for the early attempts to render boids in pygame and incorporated the “laggy predator” design. He worked to debug the serial read problems, implementing several versions of serial read in Python including versions with manual buffer management in the effort to circumvent problems with the readline() method. He collected data to troubleshoot the communication problem and produced all of the measurements and plots shown in the Early Communications Attempts section. He worked with Karim and Sonia to build and test communication options 2 and 3, and he assembled the breadboard and wood block mount for the PIC and the Pi and wired the breadboard for serial communication. He collaborated with Sonia to write the game code and optimize the game performance.